ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** -6	$a_n = a + (n-1)d$
2	**(d)** 6	BPT
3	**(d)** 5	Distance formula
4	**(b)** 84	Reverse AP has difference $-d$
5	**(c)** 9	Sides in ratio $\sqrt{\text{areas}}$
6	**(a)** $(2,\ 3)$	Section formula
7	**(c)** 3	$a_{n+1} - a_n = 3$
8	**(a)** 18	$d^2 = s^2 + s^2$
9	**(c)** 12	$\frac{1}{2} \times$ base $\times$ height
10	**(a)** 3	$a_1 = S_1$
11	**(d)** 25	Pythagoras
12	**(a)** 8	$x = 2m - a$
13	**(c)** 36	$S_n = n^2$
14	**(b)** right-angled	Check $9^2 + 12^2$ against 15^2
15	**(b)** 5	Distance from y-axis $= \lvert x \rvert$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$a_n = 3 + (n-1)(5)$ (1); $n = 13$ (1).	1+1
17	$d^2 = 289 - 225 = 64$ (1); $d = 8$ m (1).	1+1
18	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(2)^2 + (-2)^2} = \sqrt{8}$ (1); $= \sqrt{8}$ units (1).	1+1+1
19	AP: $3, 6, 9, \ldots$ (1); $S_n = \frac{10}{2}[6 + 27] = 165$ (1).	1+1
20	$25^2 = 625$; sum of squares of others $= 625$ (1); equal, so right-angled (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$a + d = 7$, $a + 4d = 19$ (2); subtracting, $d = 4$, $a = 3$ (2); $a_{10} = 39$ (1).	2+2+1
22	By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{4}{6} = \frac{12}{EC}$ (1); $EC = 18$ cm (1).	1+1+1
23	Area of triangle must be 0 (1); $\frac{1}{2}\lvert 1(6-k) + 3(k-2) + 5(2-6) \rvert = 0$ (1); $k = 10$ (1).	1+1+1
24	$a = 18$, $d = -\frac{5}{2}$, $l = -\frac{19}{2}$ (1); $l = a + (n-1)d$ gives $n - 1 = 11$ (1); $n = 12$ (1).	1+1+1
	OR: Ratio of areas $=$ (ratio of sides)^2 (1); $\frac{16}{36} = \frac{4^2}{x^2}$ (1); $x = 6$ cm (1).	1+1+1
25	Area $= \frac{1}{2}\lvert x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \rvert$ (1); $= \frac{1}{2}\lvert 3 + 4 - 28 \rvert$ (1); $= \frac{21}{2}$ sq. units (1).	1+1+1
	OR: $a = 15$, $d = 4$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{22} = \frac{22}{2}[30 + 21(4)] = 1254$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$S_n = \frac{n}{2}[2(24) + (n-1)(-2)] = 144$ (1); form quadratic in n (2); $n = 9$ (the other root is rejected) (2).	1+2+2
	OR: Given $\triangle ABC$, $DE \parallel BC$; join BE, CD and draw $EM \perp AB$, $DN \perp AC$ (1); $\frac{ar(ADE)}{ar(BDE)} = \frac{\frac{1}{2}AD \cdot EM}{\frac{1}{2}DB \cdot EM} = \frac{AD}{DB}$ and $\frac{ar(ADE)}{ar(CDE)} = \frac{AE}{EC}$ (2); $ar(BDE) = ar(CDE)$ (same base DE, same parallels) (1); hence $\frac{AD}{DB} = \frac{AE}{EC}$ (1).	1+2+1+1
27	Let the point be $(x, 0)$; $(x-7)^2 + 36 = (x+3)^2 + 16$ (2); $x = 3$ (2); point $(3, 0)$ (1).	2+2+1
	OR: AP: $a = 100$, $d = 10$, $n = 20$ (1); last $= a + (n-1)d = 290$ (2); total $= S_n = \frac{20}{2}(100 + 290) = 3900$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME — SET 2

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 5	$a + 9d$
2	**(b)** 5	$\frac{AE}{AC} = \frac{AD}{AB}$
3	**(b)** 3	Equate the squared distances
4	**(b)** 78	$a_n = a + (n-1)d$
5	**(d)** 12	BPT
6	**(b)** 13	Distance formula
7	**(c)** 59	Reverse AP has difference $-d$
8	**(b)** 12	Sides in ratio $\sqrt{\text{areas}}$
9	**(d)** $(5,\ 6)$	Section formula
10	**(d)** 3	$a_{n+1} - a_n = 3$
11	**(b)** 50	$d^2 = s^2 + s^2$
12	**(b)** 20	$\frac{1}{2} \times$ base $\times$ height
13	**(a)** 4	$a_1 = S_1$
14	**(b)** 40	Pythagoras
15	**(a)** 10	$x = 2m - a$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{3(8)+1(4)}{4} = 7$ (1); $y = 3$ (1).	1+1+1
17	$n = 12$ (1); $S = \frac{12}{2}(-5 + 28) = 138$ (1).	1+1
18	Side ratio $= 1:2$ (1); area ratio $= 1:4$ (1).	1+1
19	$(8)^2 + (y+3)^2 = 100$ (1); $(y+3)^2 = 36$ (1); $y = -3 \pm 6$, i.e. $y = 3$ or -9 (1).	1+1+1
20	$a_n = 8 + (n-1)(3)$ (1); $n = 22$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$\triangle ADE \sim \triangle ABC$, $AD : AB = 1:3$ (1); area ratio $= 1:9$ (2); trapezium area $= 8$ parts, so $\triangle ADE : DBCE = 1:8$ (2).	1+2+2
22	Let ratio $m:n$; $y = \frac{m(8)+n(-6)}{m+n} = 0$ (2); $m:n = \frac{3}{4}:1$, i.e. 3:4 (1); point $x = \frac{3(3)+4(-4)}{7} = -1$ (2).	2+1+2
23	$a + 2d = 11$, $a + 6d = 23$ (2); subtracting, $d = 3$, $a = 5$ (2); $a_{12} = 38$ (1).	2+2+1
24	By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{5}{3} = \frac{10}{EC}$ (1); $EC = 6$ cm (1).	1+1+1
	OR: Area of triangle must be 0 (1); $\frac{1}{2}\lvert 2(5-k) + 4(k-1) + 6(1-5)\rvert = 0$ (1); $k = 9$ (1).	1+1+1
25	$a = 6$, $d = 4$, $l = 86$ (1); $l = a + (n-1)d$ gives $n - 1 = 20$ (1); $n = 21$ (1).	1+1+1
	OR: Ratio of areas $=$ (ratio of sides)^2 (1); $\frac{18}{162} = \frac{3^2}{x^2}$ (1); $x = 9$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Ratio of sides $=$ ratio of perimeters $= 1:2$ (1); side $= 10$ cm (1); ratio of areas $= 1:4$ (1).	1+1+1
	OR: Diagonals bisect each other: midpoint of AC is $\left(3, \frac{9}{2}\right)$ (2); this is also the midpoint of BD (1); $D = (2,\ 6)$ (2).	2+1+2
27	$S_n = \frac{n}{2}[2(9) + (n-1)(8)] = 450$ (1); form quadratic in n (2); $n = 10$ (the other root is rejected) (2).	1+2+2
	OR: If $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$ (1). Suppose not; draw $DE' \parallel BC$ meeting AC at E' (1); by BPT $\frac{AD}{DB} = \frac{AE'}{E'C}$ (1); so $\frac{AE}{EC} = \frac{AE'}{E'C}$, hence $E = E'$ (1); thus $DE \parallel BC$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 64	$S_n = n^2$
2	**(b)** not right-angled	Check $7^2 + 8^2$ against 10^2
3	**(c)** 7	Distance from y-axis $= \lvert x \rvert$
4	**(d)** -20	$a + 9d$
5	**(d)** 8	$\frac{AE}{AC} = \frac{AD}{AB}$
6	**(a)** $\frac{3}{2}$	Equate the squared distances
7	**(a)** 17	$a_n = a + (n-1)d$
8	**(d)** 12	BPT
9	**(c)** 15	Distance formula
10	**(d)** 125	Reverse AP has difference $-d$
11	**(d)** 6	Sides in ratio $\sqrt{\text{areas}}$
12	**(a)** $(0, 2)$	Section formula
13	**(d)** 3	$a_{n+1} - a_n = 3$
14	**(d)** 72	$d^2 = s^2 + s^2$
15	**(a)** 15	$\frac{1}{2} \times$ base $\times$ height

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$d^2 = 625 - 576 = 49$ (1); $d = 7$ m (1).	1+1
17	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(4)^2 + (-4)^2} = \sqrt{32}$ (1); $= \sqrt{32}$ units (1).	1+1+1
18	AP: $5, 10, 15, \ldots$ (1); $S_n = \frac{12}{2}[10 + 55] = 390$ (1).	1+1
19	$10^2 = 100$; sum of squares of others $= 85$ (1); not equal, so not right-angled (1).	1+1
20	Section formula $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$ (1); $x = \frac{2(4)+3(-1)}{5} = 1$ (1); $y = 3$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Area $= \frac{1}{2}\lvert x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \rvert$ (1); $= \frac{1}{2}\lvert 8 + 7 + 6 \rvert$ (1); $= \frac{21}{2}$ sq. units (1).	1+1+1
22	$a = -9$, $d = 2$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{25} = \frac{25}{2}[-18 + 24(2)] = 375$ (1).	1+1+1
23	$\triangle ADE \sim \triangle ABC$, $AD : AB = 2 : 5$ (1); area ratio $= 4 : 25$ (2); trapezium area $= 21$ parts, so $\triangle ADE : DBCE = 4 : 21$ (2).	1+2+2
24	Let ratio $m : n$; $y = \frac{m(9)+n(-3)}{m+n} = 0$ (2); $m : n = \frac{1}{3} : 1$, i.e. 1:3 (1); point $x = \frac{1(6)+3(2)}{4} = 3$ (2).	2+1+2
	OR: $a + 3d = 16$, $a + 5d = 28$ (2); subtracting, $d = 6$, $a = -2$ (2); $a_9 = 46$ (1).	2+2+1
25	By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{2}{7} = \frac{10}{EC}$ (1); $EC = 35$ cm (1).	1+1+1
	OR: Area of triangle must be 0 (1); $\frac{1}{2}\lvert -1(7-k) + 1(k-3) + 3(3-7) \rvert = 0$ (1); $k = 11$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Let the point be $(x, 0)$; $(x-2)^2 + 9 = (x-8)^2 + 1$ (2); $x = \frac{13}{3}$ (2); point $(\frac{13}{3}, 0)$ (1).	2+2+1
	OR: AP: $a = 200$, $d = 15$, $n = 12$ (1); last $= a + (n-1)d = 365$ (2); total $= S_n = \frac{12}{2}(200 + 365) = 3390$ (2).	1+2+2
27	Ratio of sides $=$ ratio of perimeters $= 1 : 3$ (1); side $= 18$ cm (1); ratio of areas $= 1 : 9$ (1).	1+1+1
	OR: Diagonals bisect each other: midpoint of AC is $\left(\frac{9}{2}, \frac{7}{2}\right)$ (2); this is also the midpoint of BD (1); $D = (3, 5)$ (2).	2+1+2

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attempted. Deduct up to ½ mark for missing units or figures where required.

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